Listening Through the Noise: Modeling a Multi-Band Equalizer for Audio Enhancement and Noise Reduction

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***Abstract*—This case study presents the design and implementation of a five-band audio equalizer using continuous-time RC-equivalent filters in MATLAB. The equalizer was built from parallel filters composed of first-order lowpass, highpass, and bandpass filters with selected cutoff frequencies at 500 Hz, 2000 Hz, 3000 Hz, and 8000 Hz. Each band was independently scaled and then recombined to modify the overall frequency content of the input audio. The system was analyzed using frequency response, phase response, and impulse response plots to verify the behavior of the individual filter bands and the combined equalizer. Three presets, Unity, Bass Boost, and Treble Boost, were applied to the music recordings 'Giant Steps' and 'Space Station' to show change in low and high frequency content. The equalizer was also applied to a field recording of bird vocalizations in order to reduce low-frequency background noise and emphasize higher-frequency bird calls. Spectrograms were used to observe how the frequency content changed over time and to identify repeating vocalization patterns.**

I. Background

Audio recordings contain signals across a wide range of frequencies. Signal processing techniques can be used to modify the frequencies in the content of an audio signal in order to enhance certain features or suppress background noise. One common approach is the use of an equalizer, which divides the frequency spectrum into several bands and allows the gain of each band to be adjusted independently. By amplifying or attenuating specific frequency ranges, equalizers can improve the clarity of recordings or emphasize certain sounds.

In systems theory, frequency-selective behavior can be implemented using filters. In this study, filters are modeled using RC circuit equivalents, which are first-order linear time-invariant (LTI) systems. A lowpass RC filter allows low-frequency signals to pass while attenuating higher frequencies, whereas a highpass RC filter suppresses low-frequency signals while allowing higher frequencies to pass. By combining lowpass and highpass filters, a bandpass filter can be constructed, which isolates signals within a specific frequency range. The filters can be analyzed through their impulse responses and frequency responses, which describe how the system responds to inputs in the time and frequency domains.

The equalizer implemented in this case study consists of multiple frequency-selective filters working in parallel. Each filter processes the input signal over a specific frequency band, and the filtered outputs are scaled by adjustable gains before being added together. The gain of each band determines whether that portion of the spectrum is amplified or attenuated.

In this project, a multi-band equalizer is designed using continuous-time filter models implemented in MATLAB. The system is then applied to music recordings and a field recording of bird vocalizations. By adjusting the gains of the different frequency bands, the equalizer can enhance musical characteristics or isolate bird calls while reducing background noise.

II. Methods

A. Equalizer Design

A five-band audio equalizer was designed using continuous-time frequency-selective filters modeled after RC circuits. Each band was implemented using first-order lowpass and highpass transfer functions. The cutoff frequency for each filter was found using the RC time constant $\tau = \frac{1}{2\pi f_c}$ where f_c is the cutoff frequency of the filter. The other cutoff frequencies were selected to divide the audio spectrum into five bands: 500 Hz, 2000 Hz, 3000 Hz, 8000 Hz, and 15000 Hz. These values were manually tuned to separate low-frequency noise from midrange and higher-frequency bird vocalizations.

The equalizer was implemented as a parallel filter bank, where the input signal was processed independently by each frequency band and then recombined. The bands are defined as:

- **Band 1:** Lowpass filter with cutoff at 500 Hz to capture low-frequency components.
- **Band 2:** Bandpass filter between 500 Hz and 2000 Hz.
- **Band 3:** Bandpass filter between 2000 Hz and 3000 Hz.
- **Band 4:** Bandpass filter between 3000 Hz and 8000 Hz, targeting many bird vocalizations.
- **Band 5:** Highpass filter above 8000 Hz representing very high-frequency components.

B. Signal Processing Implementation

All signals were processed in MATLAB using the lsim() function to simulate the response of each continuous-time filter to the input audio signal. The audio recordings were first imported using audioread, and a corresponding time vector was generated using the sampling frequency.

For each band i, the filtered output $y_i(t)$ was obtained by

$$y_i(t) = x(t) * h_i(t)$$

where $x(t)$ is the input signal and $h_i(t)$ is the impulse response of the corresponding filter. The outputs of the five bands were then multiplied by adjustable gains and summed to produce the final equalized signal

$$y(t) = \sum_{i=1}^{5} g_i y_i(t)$$

where g_i represents the gain applied to band i. The signal was then normalized to prevent clipping during playback.

C. Preset Equalizer Configurations

Three preset equalizer configurations were implemented to show different spectral modifications:

- **Unity Preset:** All gains set to 1 to approximate a flat response.
- **Bass Boost:** Gains for Bands 1 and 2 increased while higher-frequency bands were attenuated.
- **Treble Boost:** Gains for Bands 4 and 5 increased while lower-frequency bands were reduced.

These presets were applied to music recordings to show how the equalizer affects the frequency balance of an audio signal.

D. Transfer Function Modeling

The filters used in the equalizer were modeled as first-order linear time-invariant systems derived from continuous-time RC circuits. Each filter is defined by a transfer function in the Laplace domain.

A first-order lowpass filter allows frequencies below the cutoff to pass while attenuating higher frequencies. Its transfer function is given by

$$H_{LP}(s) = \frac{1/\tau}{s + 1/\tau} \tag{1}$$

where τ is the RC time constant.

A first-order highpass filter attenuates low-frequency signals while allowing higher frequencies to pass and is defined by

$$H_{HP}(s) = \frac{s}{s + 1/\tau}. \tag{2}$$

The intermediate equalizer bands are implemented as bandpass filters by cascading a highpass and a lowpass filter. In the Laplace domain, the resulting transfer function is the product of the individual components,

$$H_{BP}(s) = \left(\frac{s}{s + 1/\tau_{HP}}\right)\left(\frac{1/\tau_{LP}}{s + 1/\tau_{LP}}\right). \tag{3}$$

In MATLAB, these transfer functions were implemented using numerator and denominator coefficient vectors. For the bandpass filters, the final polynomial coefficients were found by convolving the numerator and denominator vectors of the highpass and lowpass components.

E. Frequency Response and Impulse Response Analysis

The frequency response of each band filter was found using the MATLAB freqs() function over a logarithmic frequency range from 10 Hz to 20 kHz. Both magnitude and phase responses were computed and plotted to show the frequency-selective behavior of each band. The total equalizer frequency response was also computed as the weighted sum of the individual band responses. As shown in Figure 1, the magnitude plots show that each band acts as a frequency filter, where lower bands isolate lower frequencies, and higher bands isolate higher frequencies.

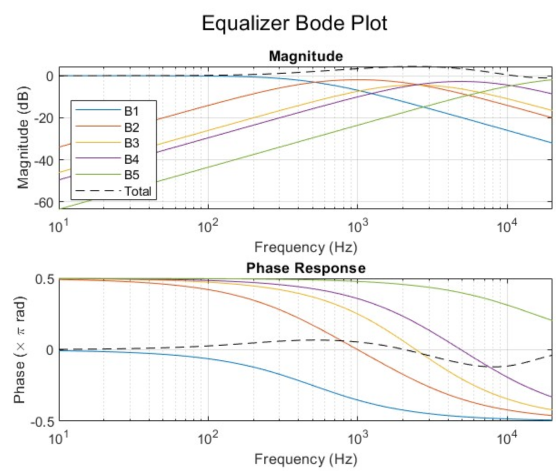

Fig. 1. Frequency response of the five equalizer bands.

The bands also exhibit smooth curves rather than short cutoffs because it depends on first-order RC filters. The dashed ”total” line shows that the overall system response is relatively flat for the unity preset, which is expected.

Impulse responses for each filter were approximated using the lsim() function by applying an impulse input of a single nonzero sample followed by zeros. This allowed visualization of the time-domain response of each filter. For stable, continuous time first-order circuits, the amplitude of the five bands decays to zero. The impulse response of the equalizer is shown in Figure 2 .

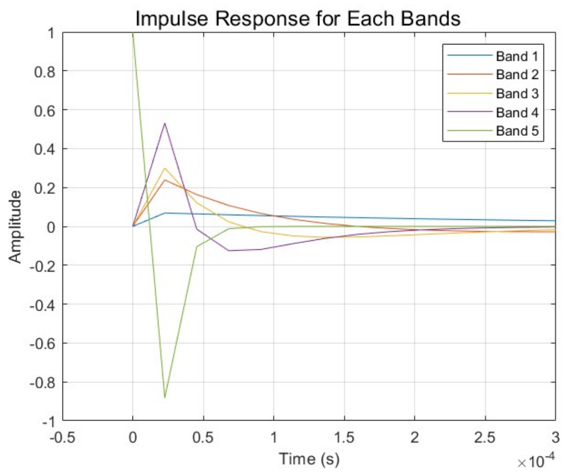

Fig. 2. Time-domain impulse response for each filter band.

As expected, Bands 4 and 5 exhibited the fastest decay because their higher cutoff frequencies correspond to smaller time constants. Bands 1–3, which have lower cutoff frequencies and therefore larger time constants, decay more gradually after the impulse.

III. Results

A. Music Recording Enhancements

Three manually tuned audio presets were used to enhance music recordings ”Giant Steps” and ”Space Station”. The band and gain settings, described above, are unity preset, bass boost, and treble boost. Visualized in the Space Station Spectogram in Figure 3 and Giant Steps Figure 4 and based on qualitative analysis, the Bass Boost amplified the low-frequency deep instruments in both tracks, whereas the Treble Boost amplified the high-frequency percussion and horn sections. It was more easily identified in the Giant Steps audio track rather than the Space Station audio, most likely due to the lack of higher frequency signals in the latter.

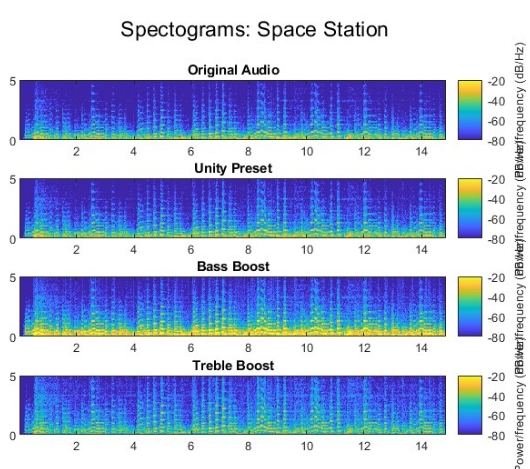

Fig. 3. Space Station Spectogram Using Audio Presets

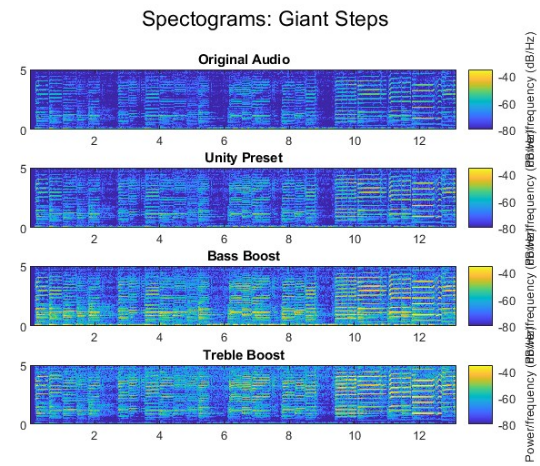

Fig. 4. Giant Steps Spectogram Using Audio Presets

Additionally, the time and frequency plots for each song and audio preset are shown in Figure 5 and Figure 6 aswell.

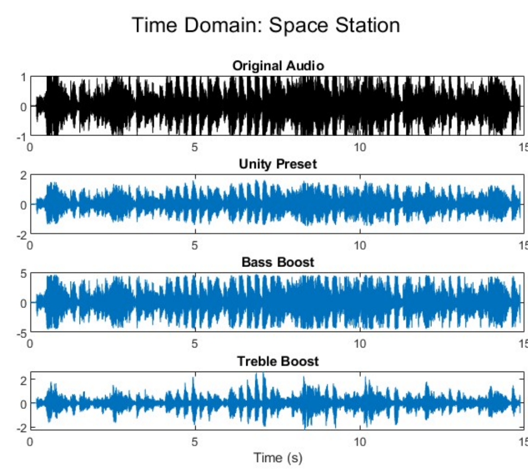

Fig. 5. Space Station Time Domain Graph

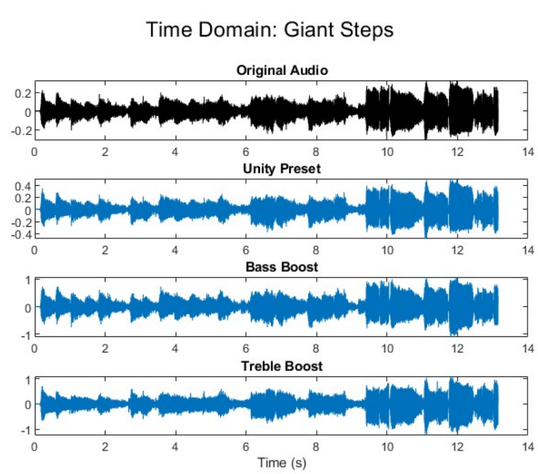

Fig. 6. Giant Steps Time Domain Graph

The plots demonstrate that the Bass Boost thickens the audio signal by shifting signal's power to lower frequencies, whereas treble sharpens the audio by focusing on higher frequency regions.

B. Bird Vocalization Analysis

The equalizer was used to process audio of various bird chirping to filter out lower-frequency wind and leaf blockage and amplify the specific bird calls. For the best visualization, the spectrogram was computed utilizing a window size of 1024 and an overlap of 800 samples. Limiting both the color axis and y-axis to a narrower range made it much easier to identify bird chirps.

Through an analysis of the spectrogram in Figure 7, 5-6 different bird species were observed, represented by the frequencies at which their chirp frequencies were registered. Several repeating bird calls were isolated, with the most prominent starting around the half-minute mark at a 3kHz frequency, likely corresponding to the Blue Jay mentioned in the prompt.

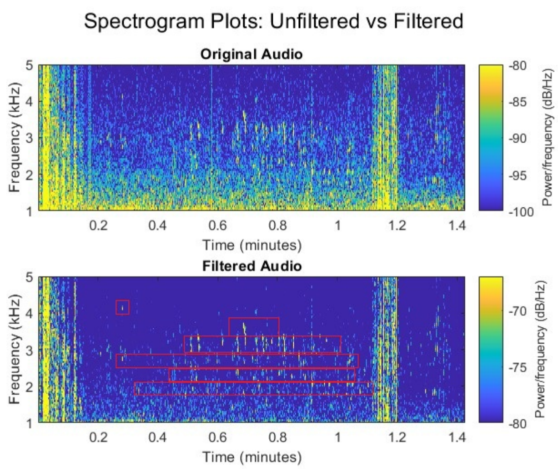

Fig. 7. Spectrogram of Filtered Bird Vocalizations

This correctly aligns with Cornell Lab's analysis of the recording. By reducing the gain at lower frequencies and increasing the gain at higher frequencies, the equalizer successfully isolated high-pitched bird localizations.

C. Design Limitations

One of the most notable design limitations was relying on simple first-order RC circuits with such wide bandwidths. Not only does this make specifying certain frequencies difficult, but the slow attenuation rate causes the bands to significantly overlap. While designing the model, we experimented on adding more bands to better specify the bird chirping frequencies, but this only caused the resulting audio to be even further blurred, most likely due to the spectral overlap. Similarly, the Bass Boost preset, when applied to audio tracks amplified signals in band 3 as well, even though gains 1 and 2 were adjusted to a high setting, and band 3 was adjusted to a very low setting.

IV. Conclusion

Overall, the equalizer met the main goal of the case study by showing that simple multi-band RC-equivalent filters can be used to modify audio recordings and improve the clarity of bird vocalizations. The results showed that adjusting gain across several frequency bands can enhance different features of a signal, whether emphasizing musical bass and treble content or reducing low-frequency background noise in environmental recordings.

The project also made the limitations of first-order filter design clear. Because the bands overlap significantly and the change between them is gradual, the system is more effective for broad enhancement rather than for precise isolation of narrow frequency components. Even with those limitations, the equalizer provided an effective application of frequency-selective filtering, and it demonstrated how signals and systems concepts can be used to analyze and improve real-world audio. Future steps that we could take to achieve sharper cut-offs is upgrade the equalizer to use a second-order RC Circuit, or even a combination of the two. This would double the attenuation rate, allowing for less overlap between the bands.